

DATABASE MANAGEMENT SYSTEM

Class :- BCA 3rd sem

Module :- 1 st

What is data and information in database?

Data are simply facts or figures — bits of **information**, but not **information** itself. When **data** are processed, interpreted, organized, structured or presented so as to make them meaningful or useful, they are called **information**. ... In above context **data** stored into **database** serve as **information**.

Data is an individual unit that contains raw material which does not carry any specific meaning.

Information is a group of **data** that collectively carry a logical meaning.

Database Elements

Tables, Records, and Fields

Tables

A database table is composed of records and fields that hold data. Tables are also called datasheets. Each table in a database holds data about a different, but related, subject.

Log ID	Operator	Resolved	Duration
1201037	CS1	☒	553
1201242	CS2	☒	524
1201247	CS1	☒	581
1201220	CS4	☐	876
1221037	CS1	☒	421

Figure 2 Database Table

Records

Data is stored in records. A record is composed of fields and contains all the data about one particular person, company, or item in a database. In this database, a record contains the data for one customer support incident report. Records appear as rows in the database table. A record for Log ID 1201242 is highlighted in Figure 3.

Log ID	Operator	Resolved	Duration
1201037	CS1	☒	553
1201242	CS2	☒	524
1201247	CS1	☒	581
1201220	CS4	☐	876
1221037	CS1	☒	421

Figure 3 Records appear as rows in a database table.

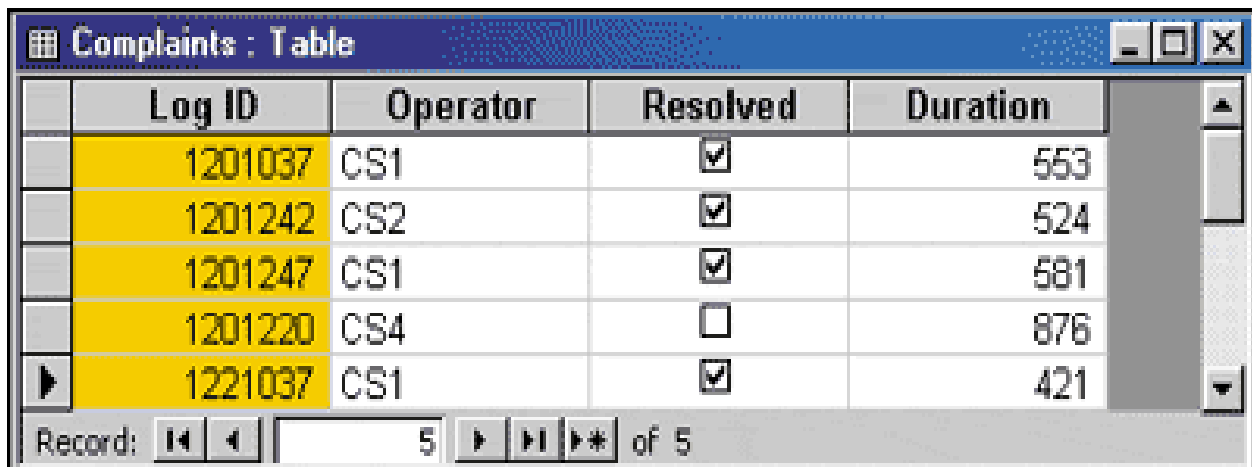
Fields

A field is part of a record and contains a single piece of data for the subject of the record. In the database table illustrated in Figure 4, each record contains four fields:

- Log ID A number assigned to this customer support incident for identification purposes
- Operator The code for the customer support operator who handled this incident
- Resolved A check box to indicate whether the incident was resolved
- Duration The time in seconds the operator spent on

this incident

Fields appear as columns in a database table. Data from the Log ID field for five records is highlighted in the Figure 4.



	Log ID	Operator	Resolved	Duration
	1201037	CS1	☒	553
	1201242	CS2	☒	524
	1201247	CS1	☒	581
	1201220	CS4	☐	876
	1221037	CS1	☒	421

Record: 5 of 5

Figure 4 Fields appear as columns in a database table.

Traditional file-based approach

The term 'file-based approach' refers to the situation where data is stored in one or more separate computer files defined and managed by different application programs. Typically, for example, the details of customers may be stored in one file, orders in another, etc. Computer programs access the stored files to perform the various tasks required by the business. Each program, or sometimes a related set of programs, is called a computer application. For example, all of the programs associated with processing customers' orders are referred to as the order processing application. The file-based approach might have application programs that deal with purchase orders, invoices, sales and marketing, suppliers, customers, employees, and so on.

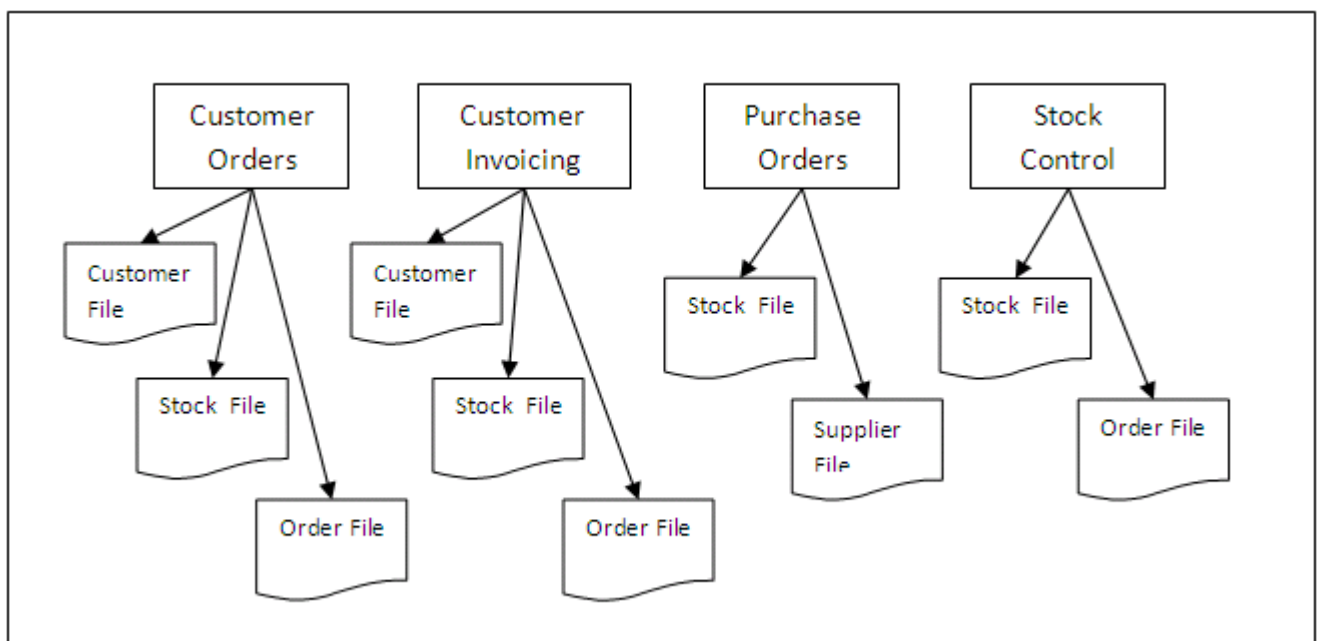
Limitations

Data duplication: Each program stores its own separate files. If the same data is to be accessed by different programs, then each program must store its own copy of the same data.

Data inconsistency: If the data is kept in different files, there could be problems when an item of data needs updating, as it will need to be updated in all the relevant files; if this is not done, the data will be inconsistent, and this could lead to errors.

Difficult to implement data security: Data is stored in different files by different application programs. This makes it difficult and expensive to implement organisation-wide security procedures on the data.

The following diagram shows how different applications will each have their own copy of the files they need in order to carry out the activities for which they are responsible:



The database approach

The database approach is an improvement on the shared file solution as the use of a database management system (DBMS) provides facilities for querying, data security and integrity, and allows simultaneous access to data by a number of different users. At this point we should explain some important terminology:

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Database: A database is a collection of related data.

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Database management system: The term 'database management system', often abbreviated to DBMS, refers to a software system used to create and manage databases. The software of such systems is complex, consisting of a number of different components, which are described later in this chapter. The term database system is usually an alternative term for database management system.

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System catalogue/Data dictionary: The description of the data in the database management system.

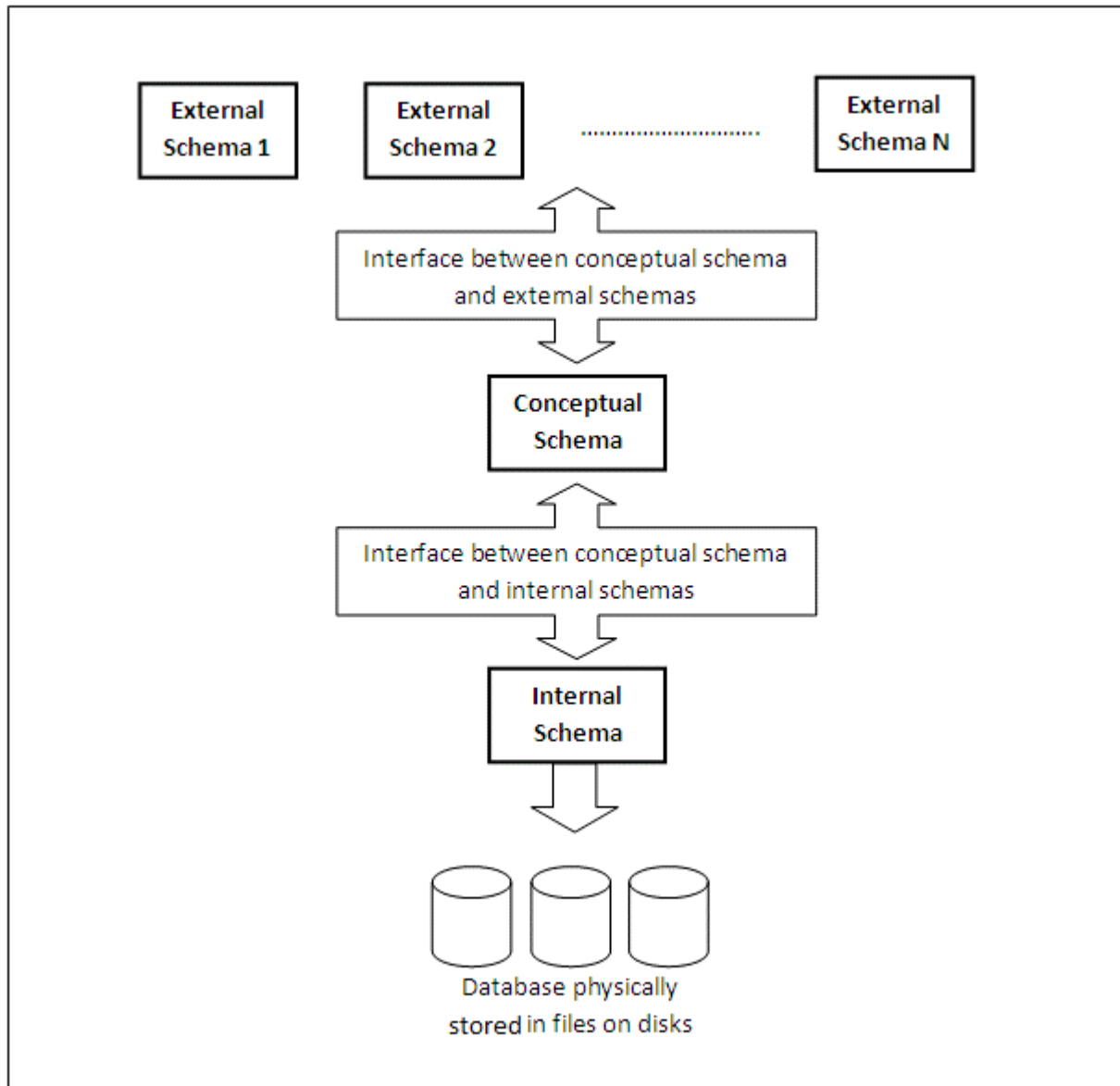
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Database application: Database application refers to a program, or related set of programs, which use the database management system to perform the computer-related tasks of a particular business function, such as order processing.

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One of the benefits of the database approach is that the problem of physical data dependence is resolved; this means that the underlying structure of a data file can be changed without the application programs needing amendment. This is achieved by a hierarchy of levels of data specification. Each such specification of data in a database system is called a schema. The different levels of schema provided in database systems are described below. Further details of what is included within each specific schema are discussed later in the chapter.

The Systems Planning and Requirements Committee of the American National Standards Institute encapsulated the concept of schema in its three-level database architecture model, known as the ANSI/SPARC architecture, which is shown in the diagram below:



What is Database

The database is a collection of inter-related data which is used to retrieve, insert and delete the data efficiently. It is also used to organize the data in the form of a table, schema, views, and reports, etc.

For example: The college Database organizes the data about the admin, staff, students and faculty etc.

Database - Advantages & Disadvantages

Advantages

- Reduced data redundancy
- Reduced updating errors and increased consistency
- Greater data integrity and independence from applications programs
- Improved data access to users through use of host and query languages
- Improved data security
- Reduced data entry, storage, and retrieval costs
- Facilitated development of new applications program

Disadvantages

- Database systems are complex, difficult, and time-consuming to design
- Substantial hardware and software start-up costs
- Damage to database affects virtually all applications programs
- Extensive conversion costs in moving from a file-based system to a database system
- Initial training required for all programmers and users

Database Components

The major components of the Database are:

• Hardware

This consists of a set of physical electronic devices such as I/O devices, storage devices and many more. It also provides an interface between computers and real-world systems.

.Software

This is the set of programs that are used to control and manage the overall Database. It also includes the DBMS software itself. The Operating System, the network software being used to share the data among the users, the application programs used to access data in the DBMS.

.Data

Database Management System collects, stores, processes, and accesses data. The Database holds both the actual or operational data and the metadata.

.Procedure

These are the rules and instructions on how to use the Database in order to design and run the DBMS, to guide the users that operate and manage it.

.Database Access Language

It is used to access the data to and from the database. In order to enter new data, updating, or retrieving requires data from databases. You can write a set of appropriate commands in the database access language, submit these to the DBMS, which then processes the data and generates it, displays a set of results into a user-readable form.

Now that you guys have understood how to create a database, let's move ahead and understand the types.

Database Management System

- Database management system is a software which is used to manage the database. For example: **MySQL**, **Oracle**, etc are a very popular commercial database which is used in different applications.
- DBMS provides an interface to perform various operations like database creation, storing data in it, updating data, creating a table in the database and a lot more.
- It provides protection and security to the database. In the case of multiple users, it also maintains data consistency.

DBMS allows users the following tasks:

- **Data Definition:** It is used for creation, modification, and removal of definition that defines the organization of data in the database.
- **Data Updation:** It is used for the insertion, modification, and deletion of the actual data in the database.
- **Data Retrieval:** It is used to retrieve the data from the database which can be used by applications for various purposes.
- **User Administration:** It is used for registering and monitoring users, maintain data integrity, enforcing data security, dealing with concurrency control, monitoring performance and recovering information corrupted by unexpected failure.

Characteristics of DBMS

- It uses a digital repository established on a server to store and manage the information.
- It can provide a clear and logical view of the process that manipulates data.
- DBMS contains automatic backup and recovery procedures.
- It contains ACID properties which maintain data in a healthy state in case of failure.
- It can reduce the complex relationship between data.
- It is used to support manipulation and processing of data.
- It is used to provide security of data.
- It can view the database from different viewpoints according to the requirements of the user.

Advantages of DBMS

- **Controls database redundancy:** It can control data redundancy because it stores all the data in one single database file and that recorded data is placed in the database.
- **Data sharing:** In DBMS, the authorized users of an organization can share the data among multiple users.
- **Easily Maintenance:** It can be easily maintainable due to the centralized nature of the database system.
- **Reduce time:** It reduces development time and maintenance need.
- **Backup:** It provides backup and recovery subsystems which create automatic backup of data from hardware and software failures and restores the data if required.

- **multiple user interface:** It provides different types of user interfaces like graphical user interfaces, application program interfaces

Disadvantages of DBMS

- **Cost of Hardware and Software:** It requires a high speed of data processor and large memory size to run DBMS software.
- **Size:** It occupies a large space of disks and large memory to run them efficiently.
- **Complexity:** Database system creates additional complexity and requirements.
- **Higher impact of failure:** Failure is highly impacted the database because in most of the organization, all the data stored in a single database and if the database is damaged due to electric failure or database corruption then the data may be lost forever.
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Functions of a DBMS

So, what does a DBMS really do? It organizes your files to give you more control over your data.

A DBMS makes it possible for users to create, edit and update data in database files. Once created, the DBMS makes it possible to store and retrieve data from those database files.

More specifically, a DBMS provides the following functions:

- **Concurrency:** concurrent access (meaning 'at the same time') to the same database by multiple users
- **Security:** security rules to determine access rights of users

- Backup and recovery: processes to back-up the data regularly and recover data if a problem occurs
- Integrity: database structure and rules improve the integrity of the data
- Data descriptions: a data dictionary provides a description of the data

Within an organization, the development of the database is typically controlled by **database administrators (DBAs)** and other specialists. This ensures the database structure is efficient and reliable.

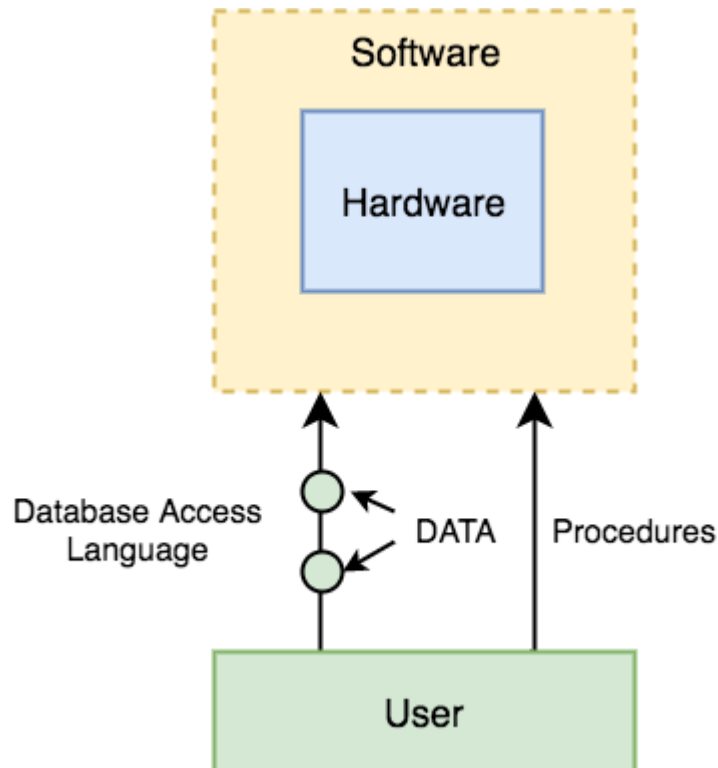
Database administrators also control access and security aspects. For example, different people within an organization use databases in different ways. Some employees may simply want to view the data and perform basic analysis. Other employees are actively involved in adding data to the database or updating existing data. This means that the database administrator needs to set the user permissions. You don't want someone who only needs to view the database to accidentally delete parts of the database.

Components of DBMS

The database management system can be divided into five major components, they are:

1. Hardware
2. Software
3. Data
4. Procedures
5. Database Access Language

Let's have a simple diagram to see how they all fit together to form a database management system.



DBMS Components: Hardware

When we say Hardware, we mean computer, hard disks, I/O channels for data, and any other physical component involved before any data is successfully stored into the memory.

When we run Oracle or MySQL on our personal computer, then our computer's Hard Disk, our Keyboard using which we type in all the commands, our computer's RAM, ROM all become a part of the DBMS hardware.

DBMS Components: Software

This is the main component, as this is the program which controls everything. The DBMS software is more like a wrapper

around the physical database, which provides us with an easy-to-use interface to store, access and update data.

The DBMS software is capable of understanding the Database Access Language and interpret it into actual database commands to execute them on the DB.

DBMS Components: Data

Data is that resource, for which DBMS was designed. The motive behind the creation of DBMS was to store and utilise data.

In a typical Database, the user saved Data is present and **meta data** is stored.

Metadata is data about the data. This is information stored by the DBMS to better understand the data stored in it.

For example: When I store my **Name** in a database, the DBMS will store when the name was stored in the database, what is the size of the name, is it stored as related data to some other data, or is it independent, all this information is metadata.

DBMS Components: Procedures

Procedures refer to general instructions to use a database management system. This includes procedures to setup and install a DBMS, To login and logout of DBMS software, to manage databases, to take backups, generating reports etc.

DBMS Components: Database Access Language

Database Access Language is a simple language designed to write commands to access, insert, update and delete data stored in any database.

A user can write commands in the Database Access Language and submit it to the DBMS for execution, which is then translated and executed by the DBMS.

User can create new databases, tables, insert data, fetch stored data, update data and delete the data using the access language.

Users

.Database Administrators: Database Administrator or DBA is the one who manages the complete database management system. DBA takes care of the security of the DBMS, its availability, managing the license keys, managing user accounts and access etc.

.Application Programmer or Software Developer: This user group is involved in developing and designing the parts of DBMS.

.End User: These days all the modern applications, web or mobile, store user data. How do you think they do it? Yes, applications are programmed in such a way that they collect user data and store the data on DBMS systems running on their server. End users are the one who store, retrieve, update and delete data.

